

Building a Coherent Proof from the Givens

Answer Key

High-School Geometry Definitions, Postulates, and Proof Logic

Quick Answers

1. 5	4. 9	7. 8	10. 8, 52
2. 5, 13	5. —	8. 22, 98, 82	
3. 12, 58	6. 35, 115	9. —	

Curriculum

Level: High-School Geometry Definitions, Postulates, and Proof Logic

HSA-REI.A.1 – *problems 1, 4*

HSG-CO.A–HSG-CO.D – *problems 2, 3, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 20 tagged checks.

1. Solve $3x + 7 = 22$ with a reason on every line.

Step 1. A justified solution is a proof: each line must be the previous line plus one named property, so undo the operations one at a time.

Step 2. $3x + 7 = 22$ (Given)

Step 3. $3x = 15$ (Subtraction Property of Equality — subtract 7 from both sides)

Step 4. $x = 5$ (Division Property of Equality — divide both sides by 3) $x = 5$

Manual-review item: the reason column is the graded part here. Accept Subtraction then Division; “inverse operations” alone is not a named property.

2. B between A and C ; $AB = 2x + 3$, $BC = 3x - 1$, $AC = 27$.

Step 1. Betweenness is what licenses adding the parts: the Segment Addition Postulate says $AB + BC = AC$. Nothing in the givens says the two parts are equal, so a sum — not an equality between the parts — is the only equation available.

Step 2. Substitute the given expressions: $(2x + 3) + (3x - 1) = 27$.

Step 3. Combine like terms: $5x + 2 = 27$, so $5x = 25$ and $x = 5$

Step 4. Substitute back into the expression for AB : $AB = 2(5) + 3 = 13$. $AB = 13$

Check: $BC = 3(5) - 1 = 14$, and $13 + 14 = 27$, the given whole.

3. $\overrightarrow{BD}$ bisects $\angle ABC$; $m\angle ABD = (2x + 5)^\circ$, $m\angle DBC = (4x - 19)^\circ$.

Step 1. The definition of an angle bisector says the two halves are congruent, and congruent angles have equal measures — that equality is the equation.

Step 2. $2x + 5 = 4x - 19$, so $24 = 2x$ and $x = 12$

Step 3. Each half measures $2(12) + 5 = 29$ degrees.

Step 4. The whole angle is the sum of the two congruent halves (Angle Addition Postulate), so $m\angle ABC = 2(29)$. $m\angle ABC = 58^\circ$

Common wrong answer: 29° — stopping at the half. The question asks for $\angle ABC$, which the bisector splits.

4. Solve $5(x - 2) = 3x + 8$ with a reason on every line.

Step 1. There are parentheses before there is anything to undo, so the first move is not a property of equality at all.

Step 2. $5(x - 2) = 3x + 8$ (Given)

Step 3. $5x - 10 = 3x + 8$ (**Distributive Property** — this is the line that is not a property of equality; it rewrites one side rather than doing the same thing to both)

Step 4. $2x - 10 = 8$ (Subtraction Property of Equality)

Step 5. $2x = 18$ (Addition Property of Equality)

Step 6. $x = 9$ (Division Property of Equality)

$$\boxed{x = 9}$$

Manual-review item: the student must name Step 3 as the Distributive Property and say that it changes only one side.

5. Given: M is the midpoint of $\overline{AB}$, N is the midpoint of $\overline{AM}$. **Prove:** $AN = \frac{1}{4}AB$.

Strategy. Each midpoint gives one halving. Apply the definition twice, then substitute one result into the other — that substitution is the step that turns two local facts into a single conclusion.

Statements	Reasons
1. M is the midpoint of $\overline{AB}$; N is the midpoint of $\overline{AM}$	1. Given
2. $AM = \frac{1}{2}AB$	2. Definition of midpoint
3. $AN = \frac{1}{2}AM$	3. Definition of midpoint
4. $AN = \frac{1}{2}(\frac{1}{2}AB)$	4. Substitution Property of Equality
5. $AN = \frac{1}{4}AB$	5. Simplify (Multiplication Property of Equality)

Manual-review item. Accept any proof that uses the midpoint definition twice and substitutes. A proof that jumps straight from the givens to $\frac{1}{4}$ has skipped the step that does the work.

6. Linear pair: $m\angle AEC = (3x + 10)^\circ$, $m\angle CEB = (x + 30)^\circ$.

Step 1. The Linear Pair Postulate is what supplies a number: two angles forming a linear pair are supplementary, so their measures sum to 180.

Step 2. $(3x + 10) + (x + 30) = 180$, so $4x + 40 = 180$ and $4x = 140$.

$$\boxed{x = 35}$$

Step 3. Substitute into the asked-for angle: $m\angle AEC = 3(35) + 10$.

$$\boxed{m\angle AEC = 115^\circ}$$

Check: $m\angle CEB = 35 + 30 = 65$, and $115 + 65 = 180$.

7. P between Q and R ; $QP = 4x - 5$, $PR = 2x + 7$, $QR = 50$. Dana wrote $4x - 5 = 2x + 7$.

Step 1 (a). Dana's equation says the two parts are equal. Nothing given says that — no congruence marks, no midpoint, no bisector. The only relationship the givens establish is that P is between Q and R , and betweenness gives a *sum*, not an equality of the parts.

Step 2 (b). The Segment Addition Postulate gives $QP + PR = QR$, so $(4x - 5) + (2x + 7) = 50$.

Step 3 (b). Combine: $6x + 2 = 50$, so $6x = 48$.

$$\boxed{x = 8}$$

Check: $QP = 27$, $PR = 23$, and $27 + 23 = 50$. Notice the two parts are genuinely unequal — which is exactly why Dana's equation could not have followed from the givens.

Manual-review item: part (a) is the graded sentence. Look for “nothing states the parts are congruent” or “betweenness gives a sum”.

8. Parallel lines: $m\angle p = (5x - 12)^\circ$, $m\angle q = (3x + 16)^\circ$, **same-side interior.**

Step 1. Same-side interior angles on parallel lines are supplementary (Same-Side Interior Angles Theorem, which needs $m \parallel n$ — without the parallel givens, no relationship exists at all).

Step 2. $(5x - 12) + (3x + 16) = 180$, so $8x + 4 = 180$ and $8x = 176$.

Step 3. $m\angle p = 5(22) - 12$.

Step 4. $m\angle q = 3(22) + 16$.

Check: $98 + 82 = 180$, as supplementary angles must.

$$x = 22$$

$$m\angle p = 98^\circ$$

$$m\angle q = 82^\circ$$

9. Given: $\overline{AB} \cong \overline{DE}$, $\overline{BC} \cong \overline{EF}$, $\angle B \cong \angle E$. **Prove:** $\overline{AC} \cong \overline{DF}$.

Strategy. $\overline{AC}$ and $\overline{DF}$ are not given directly, so they have to arrive as corresponding parts — which means proving the whole triangles congruent first. The given angle sits between the two given sides, so SAS is the postulate that fits.

Statements	Reasons
1. $\overline{AB} \cong \overline{DE}$	1. Given
2. $\angle B \cong \angle E$	2. Given
3. $\overline{BC} \cong \overline{EF}$	3. Given
4. $\triangle ABC \cong \triangle DEF$	4. SAS Congruence Postulate
5. $\overline{AC} \cong \overline{DF}$	5. CPCTC

Manual-review item. The order of statements 1–3 does not matter, but the angle must be identified as the *included* angle for SAS to apply, and step 5 must cite CPCTC rather than restating the given.

10. Synthesis: $\angle ABD \cong \angle DBC$, $m\angle ABC = (7x - 4)^\circ$, $m\angle ABD = (2x + 10)^\circ$.

Step 1 (a). Model argument. $\overline{BD}$ lies inside $\angle ABC$, so by the Angle Addition Postulate $m\angle ABD + m\angle DBC = m\angle ABC$. It is given that $\angle ABD \cong \angle DBC$, and congruent angles have equal measures, so $m\angle DBC = m\angle ABD$. Substituting that into the first equation gives $m\angle ABD + m\angle ABD = m\angle ABC$, that is, $2 \cdot m\angle ABD = m\angle ABC$. Replacing each measure by its given expression yields $7x - 4 = 2(2x + 10)$.

Step 2 (b). Distribute the right side: $7x - 4 = 4x + 20$.

Step 3 (b). Subtract $4x$ and add 4 to both sides: $3x = 24$.

Step 4 (b). Substitute into the whole angle: $m\angle ABC = 7(8) - 4$.

Check: each half measures $2(8) + 10 = 26$ degrees, and $26 + 26 = 52$.

Manual-review item: part (a) is graded on the chain, not the answer. The three named moves — Angle Addition Postulate, congruent angles have equal measures, Substitution — must appear in an order that actually reaches the equation.

$$x = 8$$

$$m\angle ABC = 52^\circ$$